

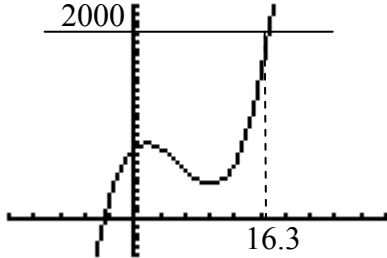
Qn	Solution
1	1st group: $2a + 0.35b + 3c = 200.88$ --- (1) 2nd group: $5a + 2b + 4.65c = 404.52$ --- (2) 3rd group: $0.9a + b + 2c = 153.12$ --- (3) Using GC, $a = 20.8, b = 36.8, c = 48.8$
2 (i)	$\frac{d}{dx} (e^{\sin \frac{x}{2}}) = \frac{1}{2} \cos \frac{x}{2} \cdot e^{\sin \frac{x}{2}}$
(ii)	$\int \sin x \cdot e^{\sin \frac{x}{2}} dx$ $= \int 2 \sin \frac{x}{2} \cos \frac{x}{2} \cdot e^{\sin \frac{x}{2}} dx$ $= 4 \sin \frac{x}{2} \cdot e^{\sin \frac{x}{2}} - \int 2 \cos \frac{x}{2} \cdot e^{\sin \frac{x}{2}} dx$ $= 4 \sin \frac{x}{2} \cdot e^{\sin \frac{x}{2}} - 4e^{\sin \frac{x}{2}} + C$
3 (i)	$z^5 - z ^4 z^* = 0$ $z^5 = z ^4 z^*$ $z^6 = z ^4 z^* z$ $= z ^4 z ^2$ $= z ^6$ $= r^6$ Since r^6 is real, z^6 is real.
(ii)	$z^6 = (r e^{i\alpha})^6 = r^6$ $e^{i6\alpha} = 1$ $e^{i6\alpha} = e^{i2k\pi} \text{ where } k = 0, \pm 1, \pm 2, 3$ $\therefore \alpha = \frac{k\pi}{3}$ Since $0 < \alpha < \frac{\pi}{2}$, $\alpha = \frac{\pi}{3}$.
3(i)	Alternatively, $z^5 - z ^4 z^* = 0$ $r^5 e^{i5\alpha} - r^4 (r e^{-i\alpha}) = 0$ $e^{i5\alpha} - e^{-i\alpha} = 0$ $e^{i6\alpha} - 1 = 0$

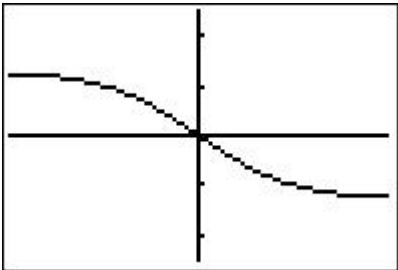
	$e^{i6\alpha} = 1$ $z^6 = r^6 e^{i6\alpha} = r^6$ Since r^6 is real, z^6 is real.
(ii)	$e^{i6\alpha} = 1$ $e^{i6\alpha} = e^{i2k\pi} \text{ where } k = 0, \pm 1, \pm 2, 3$ $\therefore \alpha = \frac{k\pi}{3}$ $0 < \alpha < \frac{\pi}{2}, \quad \alpha = \frac{\pi}{3}$ Since Alternatively for (ii), $0 < \alpha < \frac{\pi}{2}, \quad 0 < 6\alpha < 3\pi$ since $\cos 6\alpha = 1$ and $\sin 6\alpha = 0$ $6\alpha = 2\pi \text{ and } 6\alpha = \pi, 2\pi$ $\alpha = \frac{\pi}{3} \text{ and } \alpha = \frac{\pi}{6}, \frac{\pi}{3}$ $\therefore \alpha = \frac{\pi}{3}$
4 (i)	
(ii)	

5	Let P_n be the statement “$u_n = \frac{2^n}{n+2} - \frac{1}{2}$”, where $n \geq 1$. When $n = 1$, LHS = $u_1 = \frac{1}{6}$ (given), $\text{RHS} = \frac{2}{3} - \frac{1}{2} = \frac{1}{6}.$ LHS = RHS, $\therefore P_1$ is true. Assume that P_k is true for some $k \geq 1$, $\text{i.e. } u_k = \frac{2^k}{k+2} - \frac{1}{2}.$ To prove: P_{k+1} is true, $\text{i.e. } u_{k+1} = \frac{2^{k+1}}{k+3} - \frac{1}{2}.$ $ \begin{aligned} u_{k+1} &= u_k + \frac{(k+1)2^k}{(k+2)(k+3)} \\ &= \frac{2^k}{k+2} - \frac{1}{2} + \frac{(k+1)2^k}{(k+2)(k+3)} \\ &= \frac{(k+3)2^k + (k+1)2^k}{(k+2)(k+3)} - \frac{1}{2} \\ &= \frac{2^k(k+3+k+1)}{(k+2)(k+3)} - \frac{1}{2} \\ &= \frac{2^k \cdot 2(k+2)}{(k+2)(k+3)} - \frac{1}{2} \\ &= \frac{2^{k+1}}{k+3} - \frac{1}{2} \end{aligned} $ $\therefore P_k$ is true $\Rightarrow P_{k+1}$ is true. Since P_1 is true and P_k is true $\Rightarrow P_{k+1}$ is true, by mathematical induction, P_n is true for all $n \geq 1$. $u_n = \frac{2^n}{n+2} - \frac{1}{2}$ $\Rightarrow 2u_n + 1 = \frac{2^{n+1}}{n+2} > 0 \quad (\because 2^{n+1} > 0, n+2 > 0 \text{ for all } n \geq 1)$
6	$y = e^{\pi x} z$ $\frac{dy}{dx} = e^{\pi x} \frac{dz}{dx} + \pi z e^{\pi x}$ $e^{\pi x} \frac{dz}{dx} + \pi z e^{\pi x} = \pi e^{\pi x} z + \sqrt{e^{2\pi x} - 3e^{2\pi x} z^2}$ $e^{\pi x} \frac{dz}{dx} + \pi z e^{\pi x} = \pi e^{\pi x} z + e^{\pi x} \sqrt{1 - 3z^2}$

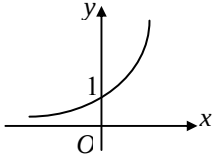
	$e^{\pi x} \frac{dz}{dx} = e^{\pi x} \sqrt{1-3z^2}$ $\frac{dz}{dx} = \sqrt{1-3z^2}$ $\int \frac{1}{\sqrt{1-3z^2}} dz = \int 1 dx$ $\frac{1}{\sqrt{3}} \sin^{-1}(\sqrt{3}z) = x + C$ When $x = 0$, $y = \frac{1}{2}$, $\Rightarrow z = \frac{1}{2} \Rightarrow C = \frac{\pi}{3\sqrt{3}} \frac{1}{\sqrt{3}} \sin^{-1}(\sqrt{3}z) = x + \frac{\pi}{3\sqrt{3}}$ $z = \frac{1}{\sqrt{3}} \sin\left(\sqrt{3}x + \frac{\pi}{3}\right)$ $y = \frac{e^{\pi x}}{\sqrt{3}} \sin\left(\sqrt{3}x + \frac{\pi}{3}\right)$
7	$(1-x)^{\frac{1}{3}} = 1 - \frac{1}{3}x + \frac{1}{2!} \cdot \frac{1}{3} \left(-\frac{2}{3}\right) (-x)^2 + \dots$ $= 1 - \frac{1}{3}x - \frac{1}{9}x^2 + \dots$ $(1+2x)^{-\frac{1}{3}} = 1 - \frac{1}{3}(2x) + \frac{1}{2!} \cdot \left(-\frac{1}{3}\right) \left(-\frac{4}{3}\right) (2x)^2 + \dots$ $= 1 - \frac{2}{3}x + \frac{8}{9}x^2 + \dots$ $\left(\frac{1-x}{1+2x}\right)^{\frac{1}{3}} = (1-x)^{\frac{1}{3}} (1+2x)^{-\frac{1}{3}}$ $= \left(1 - \frac{1}{3}x - \frac{1}{9}x^2 + \dots\right) \left(1 - \frac{2}{3}x + \frac{8}{9}x^2 + \dots\right)$ $= 1 - x + x^2 + \dots$ For expansion to be valid, $ -x < 1 \quad 2x < 1$ and $ x < 1 \quad x < \frac{1}{2}$ $\therefore x < \frac{1}{2} \text{ or } \Rightarrow -\frac{1}{2} < x < \frac{1}{2}$ Both values of x are within the range of values of x. $x = \frac{1}{6}$ will give a better approximation as it is closer to zero. Subst $x = \frac{1}{6}$,

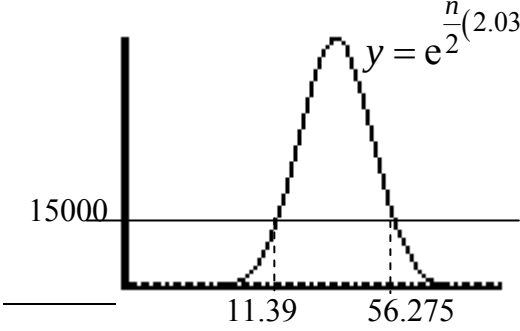
	$\left(\frac{1-\frac{1}{6}}{1+\frac{1}{6}}\right)^{\frac{1}{3}} = \left(\frac{\frac{5}{6}}{\frac{7}{6}}\right)^{\frac{1}{3}}$ $= \left(\frac{5}{7}\right)^{\frac{1}{3}}$ $\left(\frac{5}{7}\right)^{\frac{1}{3}} = 1 - \left(\frac{1}{6}\right) + \left(\frac{1}{6}\right)^2 + \dots$ $(5)^{\frac{1}{3}} \approx 2 \left(1 - \frac{1}{6} + \frac{1}{36}\right)$ $\approx \frac{31}{18}$
8	$u_{n+1} = u_n + 2n^2 - 5n$ $u_1 = u_0$ $u_2 = u_1 + 2(1)^2 - 5(1) = u_0 + 2(1)^2 - 5(1)$ $u_3 = u_2 + 2(2)^2 - 5(2) = u_0 + 2(1^2 + 2^2) - 5(1+2)$ $u_4 = u_3 + 2(3)^2 - 5(3) = u_0 + 2(1^2 + 2^2 + 3^2) - 5(1+2+3)$ $u_n = u_0 + 2(1^2 + 2^2 + 3^2 + \dots + (n-1)^2)$ $- 5(1+2+3+\dots+(n-1))$ $= u_0 + 2 \sum_{r=1}^{n-1} r^2 - 5 \sum_{r=1}^{n-1} r$ $= u_0 + 2 \cdot \frac{1}{6} (n-1)n(2n-1) - 5 \frac{(n-1)}{2} (1+n-1)$ $= u_0 + \frac{1}{6} n(n-1)(4n-17)$ $k = \frac{1}{6}$ <u>Alternative method</u> $u_{n+1} - u_n = 2n^2 - 5n$ $\sum_{r=0}^{n-1} (u_{r+1} - u_r) = \sum_{r=0}^{n-1} (2r^2 - 5r)$ $2 \sum_{r=1}^{n-1} r^2 - 5 \sum_{r=1}^{n-1} r = u_1 - u_0$ $+ u_2 - u_1$ $+ u_3 - u_2$ $+ \dots$ $+ u_n - u_{n-1}$ $= u_n - u_0$ $u_n - u_0 = 2 \cdot \frac{1}{6} (n-1)n(2n-1) - 5 \frac{(n-1)}{2} (1+n-1)$

	$= \frac{1}{6}n(n-1)(4n-17)$ $u_n = u_0 + \frac{1}{6}n(n-1)(4n-17), \quad k = \frac{1}{6}$ $u_0 = 10$ $u_n = 10 + \frac{1}{6}n(n-1)(4n-17)$ From GC, $u_{16} = 1890 < 2000$ $u_{17} = 2322 > 2000$ $\therefore$ it takes 17 days for the number of bacteria to first exceed 2 000. <u>Alternatively,</u> $u_n = 10 + \frac{1}{6}n(n-1)(4n-17) > 2000$  $n > 16.3$ Required no. of days = 17
9	$y = \sin[\ln(1-3x)]$ $\frac{dy}{dx} = \cos[\ln(1-3x)] \left(\frac{-3}{1-3x} \right)$ $(1-3x) \frac{dy}{dx} = -3 \cos[\ln(1-3x)]$ Diff. wrt x $(1-3x) \frac{d^2y}{dx^2} - 3 \frac{dy}{dx} = 3 \sin[\ln(1-3x)] \left(\frac{-3}{1-3x} \right)$ $(1-3x)^2 \frac{d^2y}{dx^2} - 3(1-3x) \frac{dy}{dx} = -9y$ $(1-3x)^2 \frac{d^2y}{dx^2} - 3(1-3x) \frac{dy}{dx} + 9y = 0 \quad (\text{shown})$ Diff. wrt x $(1-3x)^2 \frac{d^3y}{dx^3} - 9(1-3x) \frac{d^2y}{dx^2} + 18 \frac{dy}{dx} = 0$ When $x = 0$, $y = 0$, $\frac{dy}{dx} = -3$, $\frac{d^2y}{dx^2} = -9$, $\frac{d^3y}{dx^3} = -27$ $\therefore y = -3x - (9) \frac{x^2}{2} - \frac{27}{3!} x^3 \dots$ $\approx -3x - \frac{9}{2} x^2 - \frac{9}{2} x^3$

	From $\frac{dy}{dx} = \cos \left[\ln(1-3x) \right] \left(\frac{-3}{1-3x} \right)$, $(1-3x) \frac{dy}{dx} = -3 \cos \left[\ln(1-3x) \right]$ $\cos \left[\ln(1-3x) \right] = -\frac{1}{3} (1-3x) \frac{dy}{dx}$ $\cos \left[\ln(1-3x) \right] \approx -\frac{1}{3} (1-3x) \left(-3 - 9x - \frac{27}{2} x^2 \right)$ $\approx 1 - \frac{9}{2} x^2$
10 (i)	$f(x) = x - 2 \tan^{-1}(2x)$ $f'(x) = 1 - \frac{4}{1+4x^2}$ $0 \leq x^2 < \frac{3}{4}$ $1 \leq 1+4x^2 < 4$ $\frac{1}{4} < \frac{1}{1+4x^2} \leq 1$ $1 < \frac{4}{1+4x^2} \leq 4$ Since $1 < \frac{4}{1+4x^2} \leq 4$, $\therefore 1 - \frac{4}{1+4x^2} < 0$, $f'(x) < 0$ So $f(x)$ decreases when x increases. (ii)  $R_f = \left(\frac{\sqrt{3}}{2} - \frac{2\pi}{3}, -\frac{\sqrt{3}}{2} + \frac{2\pi}{3} \right)$ Or $R_f = (-1.23, 1.23)$
(iii)	For ff to exist, $R_f \subseteq D_f$ $R_f = \left(\frac{\sqrt{3}}{2} - \frac{2\pi}{3}, -\frac{\sqrt{3}}{2} + \frac{2\pi}{3} \right)$ or $R_f = (-1.23, 1.23)$ $D_f = \left(-\frac{\sqrt{3}}{2}, \frac{\sqrt{3}}{2} \right)$ or $R_g = (-0.866, 0.866)$ $\therefore R_f \not\subseteq D_f$ ff does not exist.
10 (iv)	Let $y = x - 2 \tan^{-1}(2x)$

	Then $x = y - 2 \tan^{-1}(2y)$
10 (b)	$h(x) = x^2 + 2$ $gh(x) = x^4 - 2x^2 + 5 = (x^2 - 1)^2 + 4$ $g(x+2) = [(x^2 + 2) - 3]^2 + 4$ $g(x) = (x-3)^2 + 4$ Alternatively, $h(x) = x^2 + 2$ $gh(x) = x^4 - 2x^2 + 5$ $= (x^2 + 2)^2 - 4x^2 - 4 - 2x^2 + 5$ $= (x^2 + 2)^2 - 6(x^2 + 2) + 13$ $\Rightarrow g(x) = x^2 - 6x + 13$
11 (i)	$(nx + y)^2 + 4(nx - y)^2 = 20 \text{ ---- (1)}$ Diff. wrt. x $2(nx + y)\left(n + \frac{dy}{dx}\right) + 8(nx - y)\left(n - \frac{dy}{dx}\right) = 0$ $\frac{dy}{dx} = \frac{n(3y - 5nx)}{(5y - 3nx)}$
11 (ii)	For tangent // y-axis, $5y - 3nx = 0$ $y = \frac{3nx}{5}$ Subst into (1) $\left(nx + \frac{3nx}{5}\right)^2 + 4\left(nx - \frac{3nx}{5}\right)^2 = 20$ $x = \pm \frac{1}{n} \left(\frac{5}{2}\right)$ When $x = \frac{5}{2n}$, $y = \frac{3}{2}$ When $x = -\frac{5}{2n}$, $y = -\frac{3}{2}$ The coordinates are $\left(\frac{5}{2n}, \frac{3}{2}\right)$ and $\left(-\frac{5}{2n}, -\frac{3}{2}\right)$
(ii)	<u>Alternatively,</u> Subst $nx = \frac{5y}{3}$ into (1), $\left(\frac{5y}{3} + y\right)^2 + 4\left(\frac{5y}{3} - y\right)^2 = 20 \text{ ... etc}$

11 (iii)	When curve cuts the x-axis, $y = 0$ $(nx)^2 + 4(nx)^2 = 20$ $x = \pm \frac{2}{n}$ For $x = \pm \frac{2}{n}$, $\frac{dy}{dx} = \frac{-5n^2 \left(\pm \frac{2}{n} \right)}{-3n \left(\pm \frac{2}{n} \right)} \quad \text{or} \quad \frac{dy}{dx} = \frac{n(-5nx)}{-3nx}$ $= \frac{5n}{3} \quad \quad \quad = \frac{5n}{3}$ Gradient of normal = $-\frac{3}{5n}$ Equation of normal at $\left(\frac{2}{n}, 0 \right)$, $y = -\frac{3}{5n}x + \frac{6}{5n^2}$ Equation of normal at $\left(-\frac{2}{n}, 0 \right)$, $y = -\frac{3}{5n}x - \frac{6}{5n^2}$
12(i)	common ratio of $G = -e^\alpha$ $ -e^\alpha = e^\alpha < 1 \text{ (since } \alpha < 0 \text{)}$ So G is convergent. 
12 (ii)	$S = \frac{e}{1+e^\alpha}$ Sum to infinity of even-numbered terms $= \frac{-e^{1+\alpha}}{1-e^{2\alpha}}$ $\frac{-e^{1+\alpha}}{1-e^{2\alpha}} = -\frac{1}{3} \left(\frac{e}{1+e^\alpha} \right)$ $\frac{e^\alpha}{1-e^\alpha} = \frac{1}{3}$ $\alpha = -\ln 4$ $= \ln \frac{1}{4}$
12 (iii)	$H: e + e^{1+\alpha} + e^{1+2\alpha} + e^{1+3\alpha} + \dots$ Product of first n terms of H $= e \cdot e^{1+\alpha} \cdot e^{1+2\alpha} \cdot e^{1+3\alpha} \dots e^{1+(n-1)\alpha}$ $= e^{\frac{n}{2}(2+(n-1)\alpha)}$ $= e^{\frac{n}{2}(2.03-0.03n)}$

	$e^{\frac{n}{2}(2.03-0.03n)} > 15000$ $\frac{n}{2}(2.03-0.03n) > \ln 15000$ $0.03n^2 - 2.03n + 2 \ln 15000 < 0$ $\Rightarrow 11.39 < n < 56.275$ Greatest value of n is 56. <u>Alternatively,</u>  $e^{\frac{n}{2}(2.03-0.03n)} > 15000$ $\Rightarrow 11.39 < n < 56.275$ Greatest value of n is 56.
13	$\overrightarrow{DQ} = 5\mathbf{i} - 3\mathbf{j} + 2\mathbf{k}$
(i)	Vector equation of line DQ, $\mathbf{r} = \overrightarrow{AD} + \lambda \overrightarrow{DQ}$ $= \begin{pmatrix} 0 \\ 3 \\ 0 \end{pmatrix} + \lambda \begin{pmatrix} 5 \\ -3 \\ 2 \end{pmatrix}$ $= 3\mathbf{j} + \lambda(5\mathbf{i} - 3\mathbf{j} + 2\mathbf{k})$ $\overrightarrow{BS} = -5\mathbf{i} + 3\mathbf{j} + 2\mathbf{k}$ Vector equation of line BS, $\mathbf{r} = \overrightarrow{AB} + \mu \overrightarrow{BS}$ $= 5\mathbf{j} + \mu(-5\mathbf{i} + 3\mathbf{j} + 2\mathbf{k})$
(ii)	Let the angle between the lines DQ and BS be θ

	$\overrightarrow{DQ} \cdot \overrightarrow{BS} = \overrightarrow{DQ} \overrightarrow{BS} \cos \theta$ $\begin{pmatrix} 5 \\ -3 \\ 2 \end{pmatrix} \cdot \begin{pmatrix} -5 \\ 3 \\ 2 \end{pmatrix} = \sqrt{38} \sqrt{38} \cos \theta$ $\theta = \cos^{-1} \left(\frac{-30}{38} \right) = 142.15^\circ$ Therefore, acute angle between the lines DQ and BS $= 180 - 142.15^\circ$ $= 37.9^\circ \text{ (to 1 dp)}$
(iii)	$\overrightarrow{BD} = \overrightarrow{BA} + \overrightarrow{AD}$ $= -5\mathbf{i} + 3\mathbf{j}$ $\overrightarrow{BR} = \overrightarrow{BC} + \overrightarrow{CR}$ $= 3\mathbf{j} + 2\mathbf{k}$ $\mathbf{n} = \begin{pmatrix} -5 \\ 3 \\ 0 \end{pmatrix} \times \begin{pmatrix} 0 \\ 3 \\ 2 \end{pmatrix} = \begin{pmatrix} 6 \\ 10 \\ -15 \end{pmatrix}$ Equation of plane π, $\mathbf{r} \cdot \begin{pmatrix} 6 \\ 10 \\ -15 \end{pmatrix} = \begin{pmatrix} 5 \\ 0 \\ 0 \end{pmatrix} \cdot \begin{pmatrix} 6 \\ 10 \\ -15 \end{pmatrix} = 30$
(iv)	$\overrightarrow{BX} = \overrightarrow{BC} + \overrightarrow{CX}$ $= 3\mathbf{j} + \frac{2}{3}\mathbf{k}$ Shortest distance from X to the plane π, $\frac{\left \begin{pmatrix} 0 \\ 3 \\ \frac{2}{3} \end{pmatrix} \cdot \begin{pmatrix} 6 \\ 10 \\ -15 \end{pmatrix} \right }{\sqrt{36 + 100 + 225}}$ $= \frac{20}{19} \text{ units}$